

We can install Mininet on a virtual machine or on native operating systems. The installation steps will change as per the platform on which you are installing it.

Mininet commands cheatsheet

1. To run the default topology, run the following command:

```
Mininet> sudo mn
```

2. To get the help of any command, use the command given below:

```
Mininet> help
```

3. To list and get the information about the nodes of the network, run the following command:

```
Mininet> nodes
```

4. To dump information about all the nodes, use the following command:

```
Mininet> dump
```

5. To display all the links of a network, run:

```
Mininet> net
```

6. To check the connectivity between host h1 and host h2, use the following command:

```
Mininet> h1 ping h2
```

7. To open two terminals each for node h1 and h2, use the command given below:

```
Mininet> xterm h1 h2
```

8. The following command creates a link, and sets the bandwidth and delay timing between host h1 and switch s2:

```
net.addLink(h1, s2, bw=20, delay="60ms")
```

9. To exit from the CLI, run the

command given below:

```
Mininet> exit
```

10. Run the following command to clean up the Mininet network configuration:

```
Mininet> sudo mn -c
```

11. The following command checks the connectivity between all the hosts present in the topology, as seen in Figure 3.

```
Mininet> pingall
```

Creating a custom topology

Mininet provides a simple Python API to create customised network topologies. We can create our customised topology by writing a few lines of Python code. This customised topology can configure the network based on the parameters that we pass into it, and this can be reused for various experiments. Figure 4 shows a

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet>
```

Figure 3: Ping testing between host h1 and host h2

topology with one controller (c0), three switches (s1, s2, s3) and six hosts (h1, h2, h3, h4, h5, h6).

The Python code (mytopo.py) given below will create the topology that is depicted in Figure 5.

```
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
def mytopo():

    nw = Mininet( controller=Controller
    )
    info( 'Add controller\n' )
    nw.addController( 'c0' )
    info( 'Add hosts\n' )
    h1 = nw.addHost( 'h1',
    ip='10.0.0.11' )
    h2 = nw.addHost( 'h2',
    ip='10.0.0.12' )
    h3 = nw.addHost( 'h3',
    ip='10.0.0.13' )
    h4 = nw.addHost( 'h4',
    ip='10.0.0.14' )
    h5 = nw.addHost( 'h5',
    ip='10.0.0.15' )
    h6 = nw.addHost( 'h6',
    ip='10.0.0.16' )
    info( 'Add switch\n' )
    s1 = nw.addSwitch( 's1' )
```

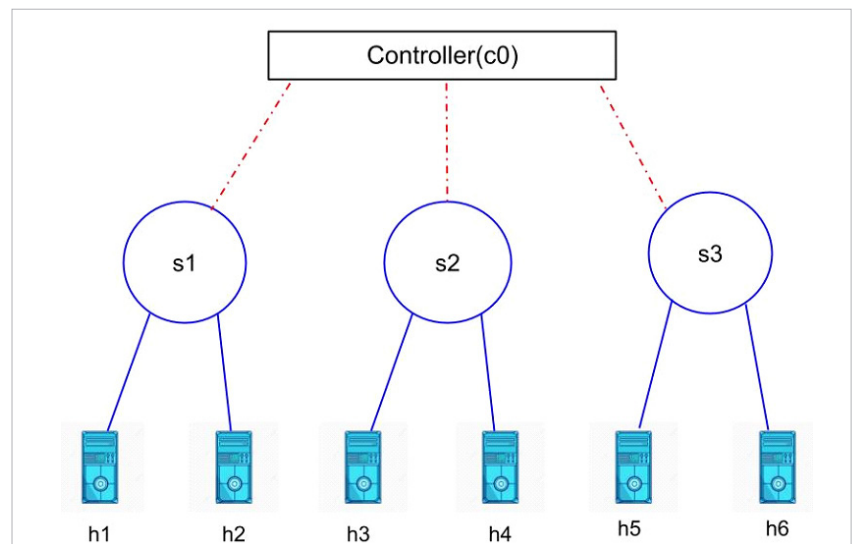


Figure 4: A sample network topology (mytopo.py)